Reddit Sentiment and Stock Returns: Evidence from AI Product Launches

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01 Motivation & Background

> *"Equity valuations appear stretched, particularly for technology companies focused on Artificial Intelligence"*
> — Bank of England, October 2025 [1]

This raises critical questions about how efficiently markets incorporate new AI products.

Theoretical Framework:

- Efficient Markets Hypothesis predicts sentiment cannot predict returns if markets are efficient [2]
- Behavioral finance suggests investors may over/underreact to social media signals, creating short-term predictability [3]
- Literature Gap: Prior work examines broad AI adoption; this study focuses on specific feature announcements at non-AI-native firms

02 Research Question

Does consumer sentiment on Reddit predict stock returns for AI feature announcements?

If it does, it suggests that markets do not immediately incorporate publicly available information, violating the Efficient Markets Hypothesis.

03 Data

Five Case Studies (2023)

Microsoft Copilot | Google Gemini | Adobe Firefly | Snapchat My AI | Spotify AI DJ

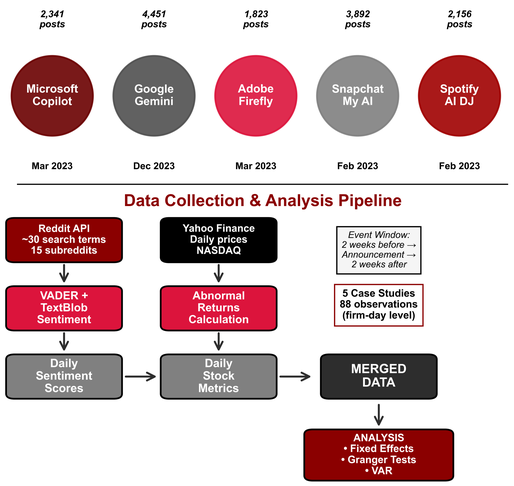
Total: 14,663 Reddit posts and comments collected.
Event Window: 2 weeks before and after each announcement (4-week window per firm).

Sentiment Data:

- Reddit API data collected across ~30 firm-related search terms and ~15 technology subreddits
- Sentiment scored using VADER and TextBlob algorithms, aggregated to daily firm-level scores

Stock Data:

- Daily stock prices from Yahoo Finance (NASDAQ)
- Abnormal returns calculated using market model with NASDAQ composite as benchmark



04 Methods

Regression Model

$$\text{Perf}_{it} = \alpha + \beta_1 \text{Sentiment}_{it} + \beta_2 \text{Perf}_{it-1} + \mu_i + \epsilon_{it}$$

Where:

- Perf_{it} = Abnormal return for firm i on day t
- Perf_{it-1} = Lagged performance (controls for momentum)
- μ_i = Firm fixed effects

I estimate the effect of lagged sentiment on abnormal returns using fixed effects regression. Coefficient β_1 measures how a one-unit increase in sentiment affects next-day returns.

Granger Causality

Establishes causal direction

- Does past sentiment predict current returns? Do past returns predict current sentiment?
- Use lags of 1 and 2 (determined by Akaike Information Criterion).

$$\text{Perf}_{it} = \alpha_0 + \sum_{k=1}^{p} \beta_k \text{Perf}_{t-k} + \sum_{k=1}^{p} \lambda_k \text{Sentiment}_{t-k} + \epsilon_t$$

$$\text{Sentiment}_t = \delta_0 + \sum_{k=1}^{p} \phi_k \text{Sentiment}_{t-k} + \sum_{k=1}^{p} \theta_k \text{Perf}_{t-k} + \nu_t$$

Joint F-tests determine whether lagged sentiment coefficients jointly equal zero. Rejecting the null means sentiment "Granger-causes" returns.

Robustness

Vector Autoregression (VAR) impulse response functions (IRFs) trace dynamic effects over a 10-day horizon.

- Shows how returns respond to a one-unit shock over time.

05 Results

Fixed-Effects Regression

Table 1: Baseline Fixed Effects Regression

	(1) Without Lag	(2) With Momentum Control
Daily Average Sentiment	-0.0032	0.0080
	(0.0130)	(0.0088)
Lagged Abnormal Returns		0.0723
		(0.0570)
Constant	0.0035	0.0007
	(0.0049)	(0.0032)

Same-day sentiment does not significantly predict same-day returns ($\beta = 0.008, p = 0.415$).

- Suggests no immediate contemporaneous relationship.

Granger Causality

Table 2: Granger Causality Test (Sentiment Predicts Returns)

	(1) Abnormal Return
L.Daily Average Sentiment	0.0216**
	(0.0064)
L2.Daily Average Sentiment	-0.0176
	(0.0146)
L.Daily Abnormal Return	-0.0007
	(0.0819)
L2.Daily Abnormal Return	-0.1700
	(0.1520)
Constant	0.0006
	(0.0059)

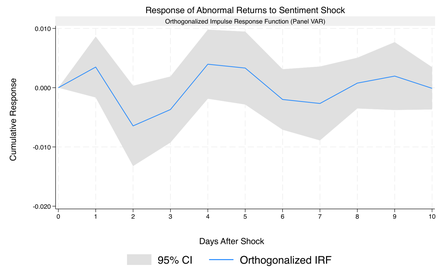
Sentiment granger-causes returns (p=0.028), but returns do not granger-cause sentiment.

- Information flows from consumer opinion to market prices, not vice versa.

06 Robustness

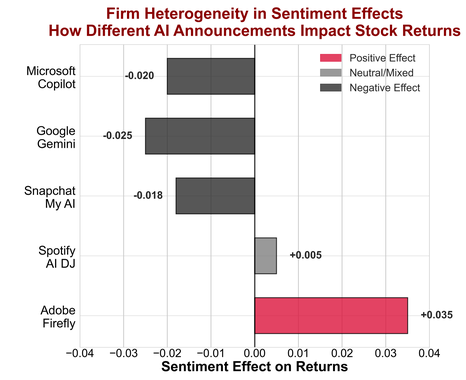
Vector Autoregression (VAR)

- VAR models how sentiment and returns predict each other over time.
- IRF traces the day-by-day response of returns to a sentiment shock.
- IRF confirms Granger results: sentiment shocks generate initial positive response followed by Day 2 overcorrection and gradual stabilization.



Firm Heterogeneity

- Markets may reward product-market fit; creative tools see a positive association.



07 Discussion

Market Efficiency

- Markets do not immediately incorporate sentiment information.
- The 1 day lag and 5-6 day dissipation (IRF) violates efficiency possibly due to:
  - Information processing delays
  - Underweighting of social media sentiment signals
  - Investor attention constraints

Implications

- Investors: Reddit sentiment offers predictive power but effects are short-lived.
- Firms: Consumer reception drives stock performance.

As the Bank of England warns of "stretched" AI valuations, understanding how markets price AI innovation becomes even more important.

References

[1] "Financial Policy Committee Record – October 2025." 2025. October 28. https://www.bankofengland.co.uk/financial-policy-committee-record/2025/october-2025.

[2] Fama, Eugene F. 1970. "Efficient Capital Markets: A Review of Theory and Empirical Work." *The Journal of Finance* 25 (2): 383–417. https://doi.org/10.2307/2325486.

[3] Barberis, Nicholas, and Richard Thaler. 2002. "A Survey of Behavioral Finance." Working Paper No. 9222. Working Paper Series. National Bureau of Economic Research, September. https://doi.org/10.3386/w9222.